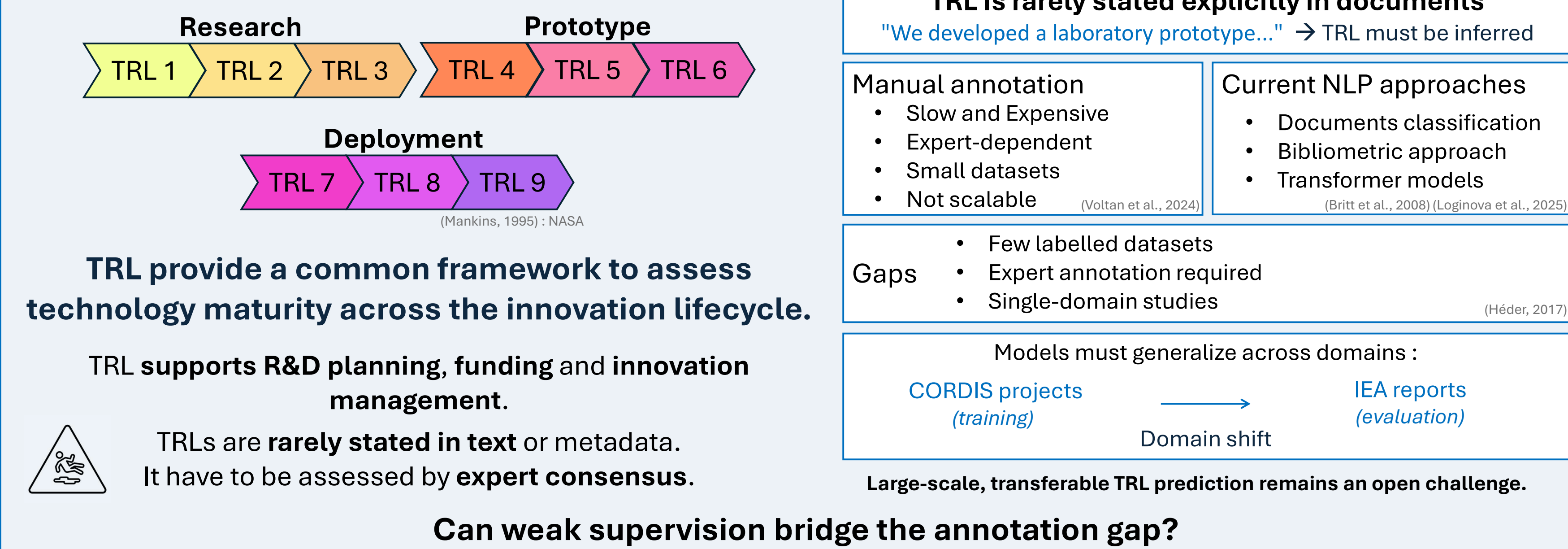




OVERVIEW

BACKGROUND & PROBLEM

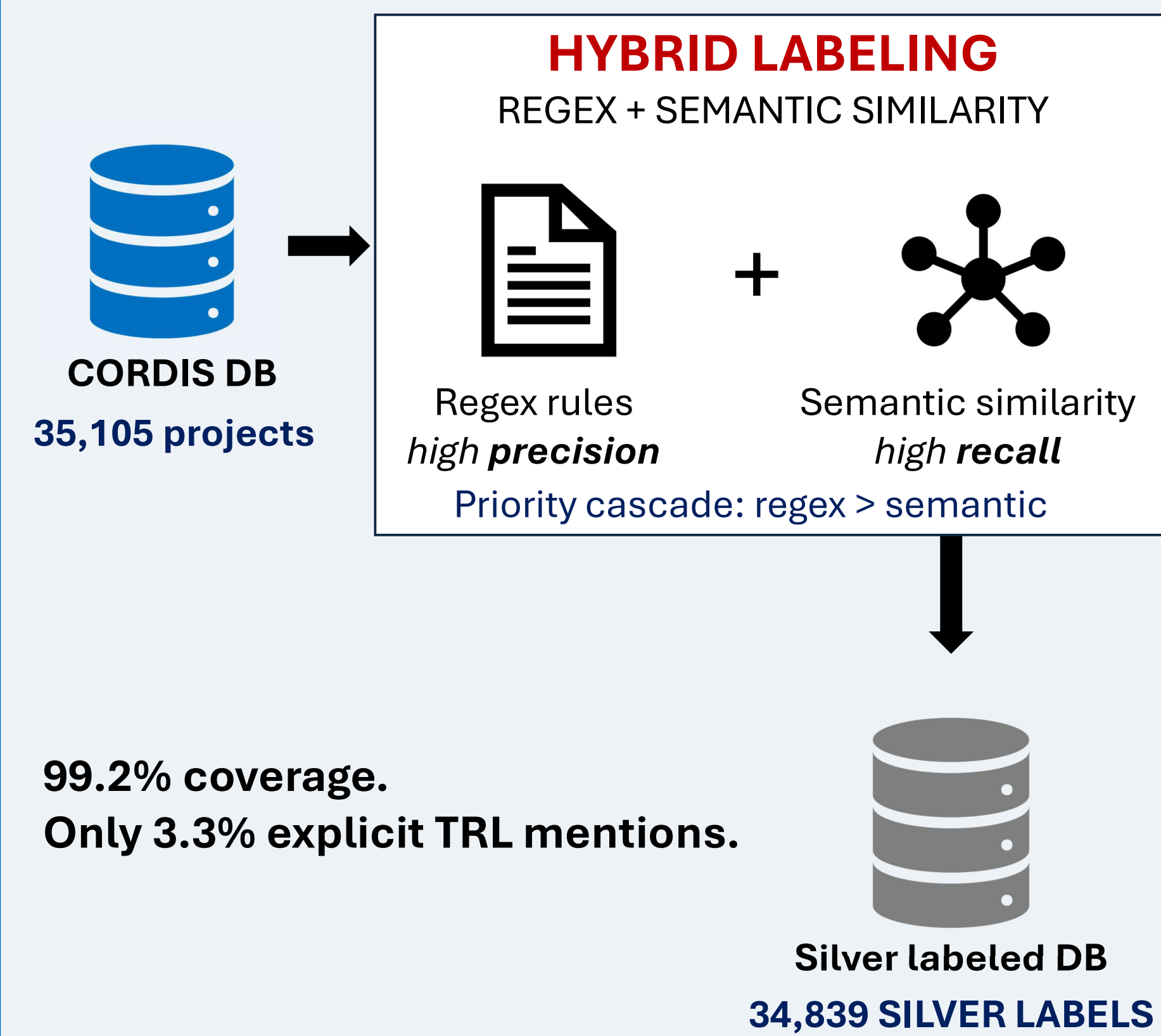


OBJECTIVES

- Generate scalable annotations
- Evaluate cross-domain transfer
- Study grammar-aware representations

METHODOLOGY

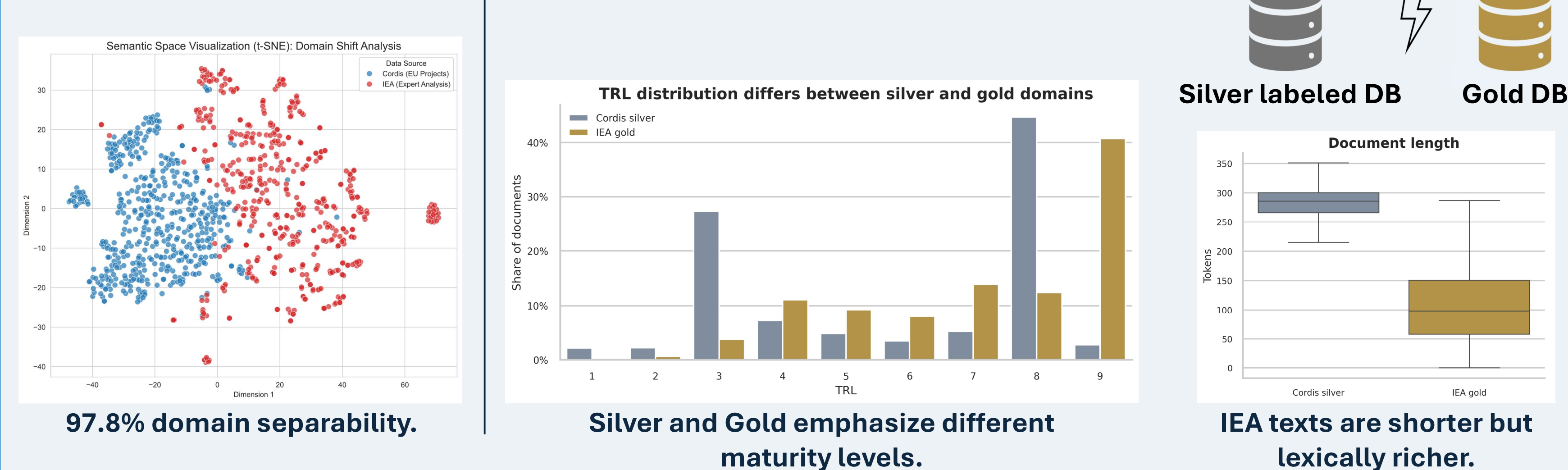
WEAK SUPERVISION



EVALUATION APPROACHES



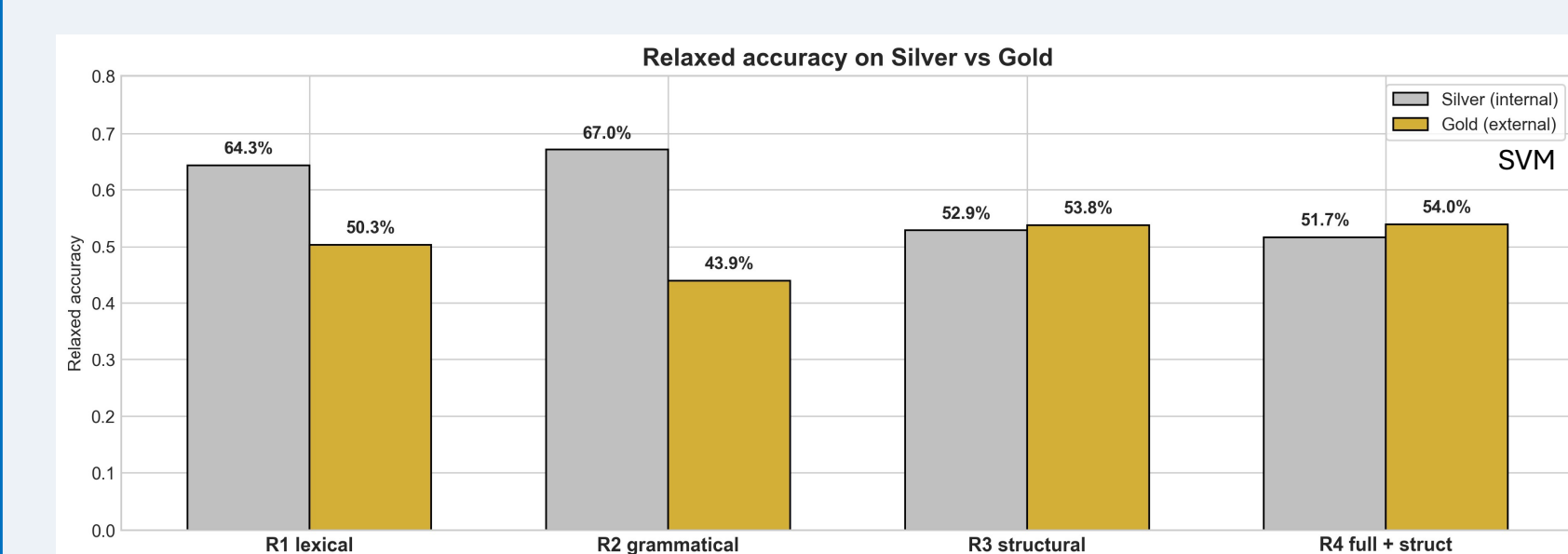
DATASETS



TEXT REPRESENTATIONS

4 hypothesis on grammar impact

Do grammatical representations generalize better than lexical representations?



Lexical representations **perform** best in-domain but **fail** to transfer.

Structural cues **improve robustness** under domain shift.

Baseline performance



R1 = Raw text

TF-IDF, words and n-gram (1-3)

'Text: "The experimental prototype..."

Grammar improves robustness



R2 = Parsed and annotated text

TF-IDF on grammar tags + TRL specific terms

'_PASSIVE__PHASE_POC_LAB__prototype'

Structural cues are sufficient



R3 = Extracted numeric features

No vectorization

'{n_tokens: 145, n_future: 0, n_modal: 2, n_passive: 1, 'has_phase_poc_lab': 1, ...}'

Fusion is optimal



R4 = R1 + R3 = raw text + features

ColumnTransformer(merge)

'Text: "The experimental prototype..." + 'Features: {n_tokens, n_passive, ...}'

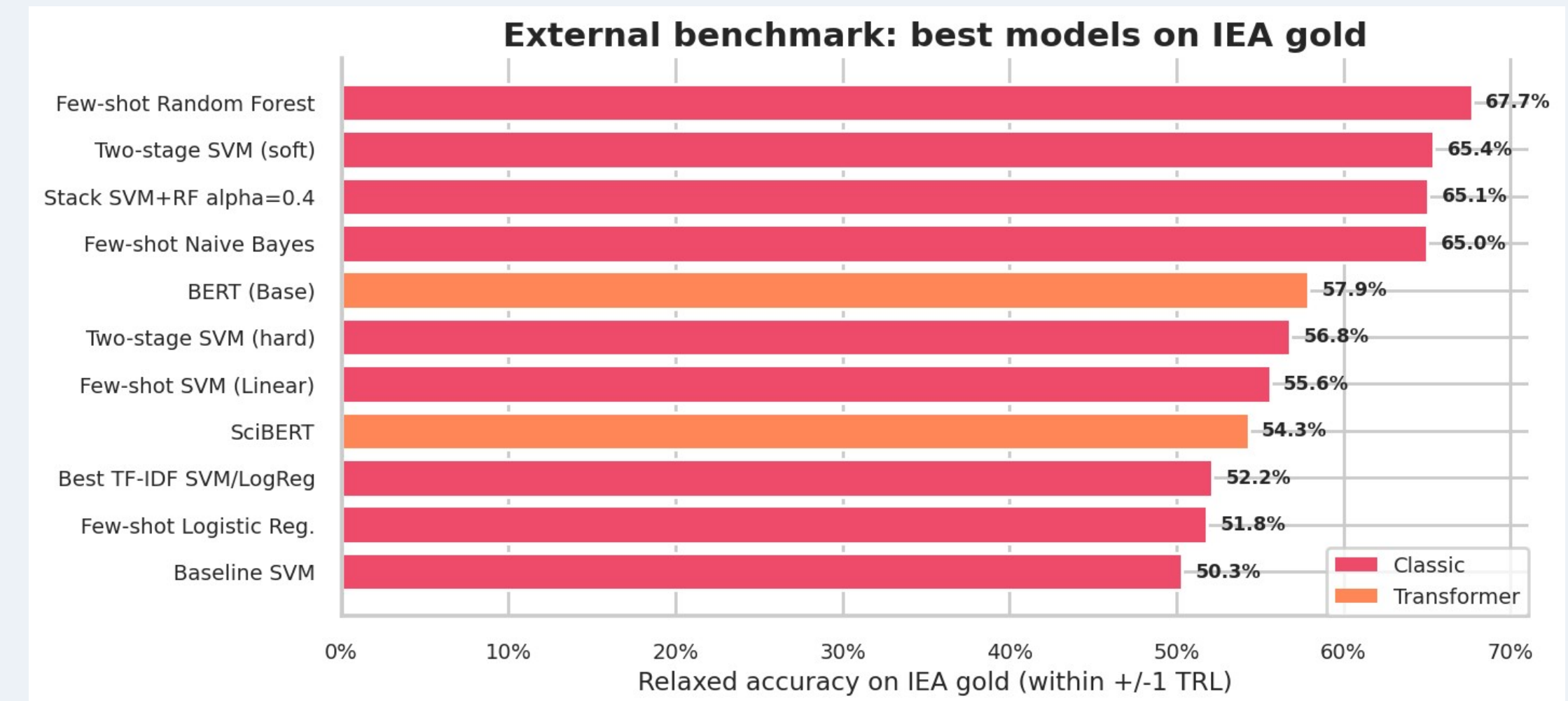
CONCLUSIONS

- Weak supervision is **scalable** for **large scale annotation**.

- Symbolic methods** challenge Transformers under **realistic low-resource conditions**.

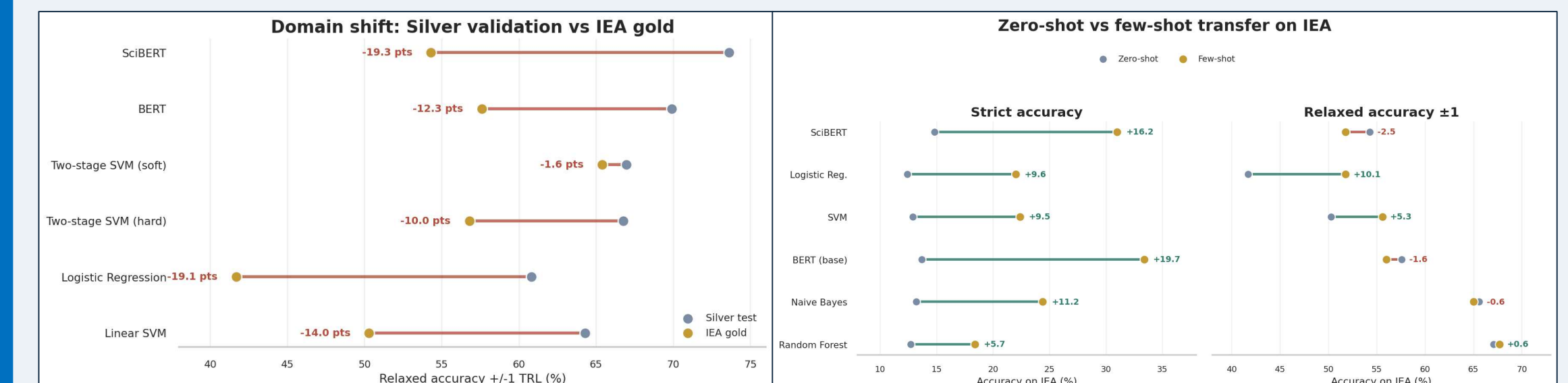
RESULTS

1. BEST MODEL



Our weakly supervised symbolic models outperform deep models on external benchmark.

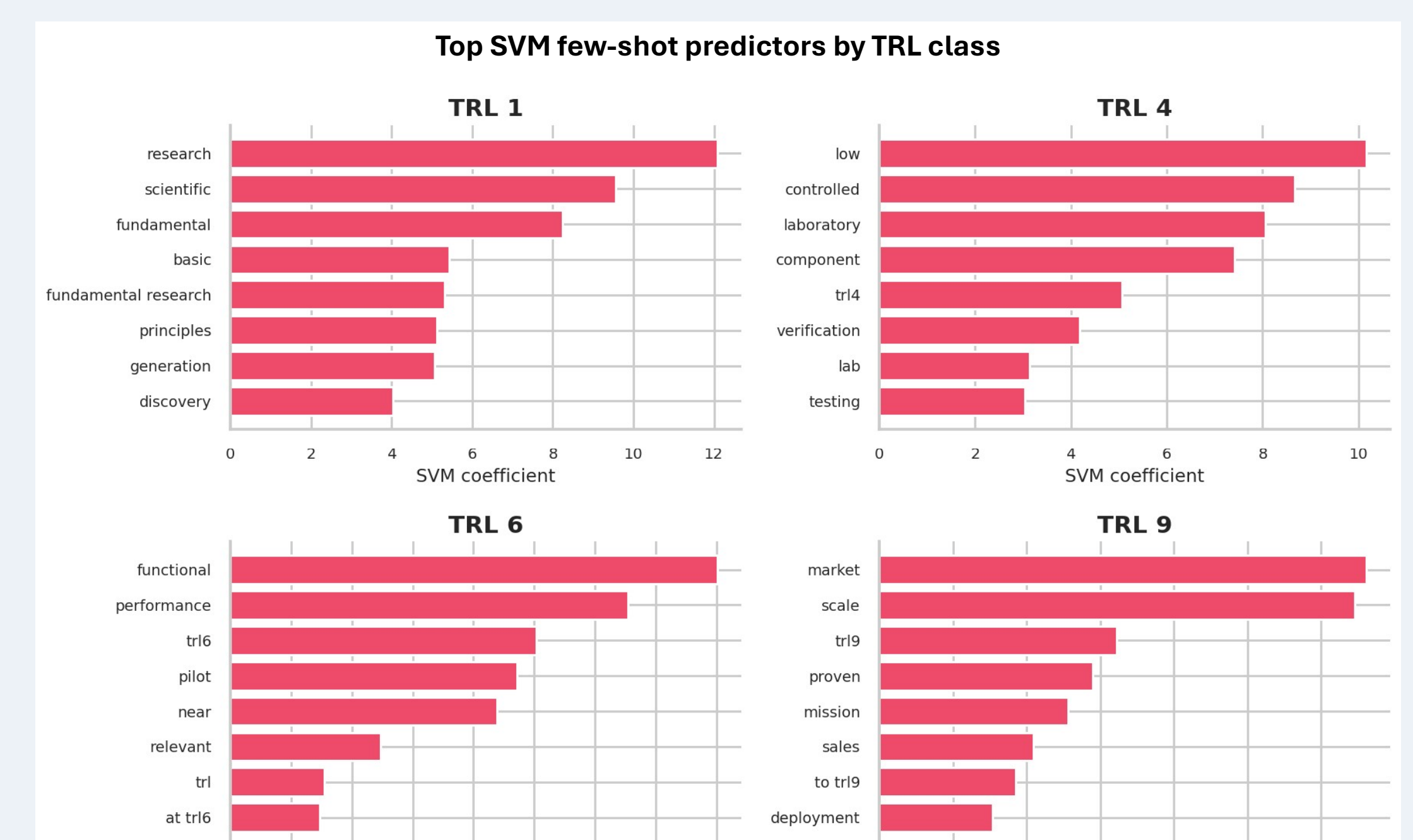
2. DOMAIN SHIFT & FEW SHOT ADAPTATION



Deep models achieve **high Silver** performance but lose up to 19 points under domain shift.

Few-shot mainly benefits classical models, gains remain limited for deep models.

3. WHAT MODEL REALLY LEARN ? EXPLAINABILITY



4. METRICS COMPARAISON

Model	Relaxed Acc Silver	Relaxed Acc Gold	MAE on Gold
SVM	0,64	0,50	2,10
Stacking (SVM+RF)	0,67	0,65	1,63
2-stage SVM (soft)	0,67	0,65	1,72
BERT (Zero-Shot)	0,70	0,58	1,92
SciBERT (Zero-Shot)	0,73	0,52	2,03

Silver validation consistently overestimates real-world performance under domain shift.

Deep learning models suffer the largest transfer degradation (~70% → ~50%).

Robust linguistic representations **generalize better** than pretrained language models in the **zero-shot** setting.

KEY FINDINGS

- ❖ 34,839 Silver TRL labels generated (**99.2% coverage**)
- ❖ TRLs exhibit a **coherent ordinal structure** (most errors within ±1 TRL)
- ❖ Semantic domain shift **limits** zero-shot **transfer**
- ❖ Grammar **improves interpretability**, not transferability

NEXT STEPS

Characterize **expert decision rules** for assigning TRLs.

Extend evaluation to additional industrial domains.

Build domain-aware **weak supervision** combining **expert knowledge**, **structured metadata**.

